

Ningzhe Shi

Ph.D. Candidate in Computer Science and Technology

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Research Profile

Ph.D. candidate researching 6G intelligent networks and AI service orchestration, with a focus on communication, sensing, computing, and knowledge resource optimization through reinforcement learning and optimization methods.

Education

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| <p>Ph.D. Institute of Computing Technology, Chinese Academy of Sciences; University of Chinese Academy of Sciences, Computer Science and Technology</p> <ul style="list-style-type: none"> • Ph.D. advisor: Prof. Yiqing Zhou • Discipline rated A+ in the national subject assessment | <p>Beijing, China
Sept 2021 – present</p> |
| <p>B.Eng. Xidian University, Communication Engineering</p> <ul style="list-style-type: none"> • Undergraduate mentor: Prof. Junyu Liu • Comprehensive recommendation ranking: 1st in class • Discipline rated A+ in the national subject assessment | <p>Xi'an, China
Sept 2017 – July 2021</p> |

Internship

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| <p>China Mobile, Digital Intelligence Department, AI Agent Algorithm Intern</p> <p>Conducted research on long-horizon memory mechanisms for AI agents.</p> <ul style="list-style-type: none"> • Studied retrieval-augmented memory, hierarchical storage, memory compression, forgetting, and dynamic updating. • Reproduced and analyzed the Mem-alpha architecture and modeled memory maintenance as a policy optimization problem. • Investigated memory-aware task scheduling in cloud-edge collaborative systems. | <p>China
Jan 2026 – Mar 2026</p> |
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Research Experience

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| <p>Institute of Computing Technology, Chinese Academy of Sciences, Technical Lead, Distributed System-Level Simulator for Large-Scale Full-Duplex Ad Hoc Networks</p> <ul style="list-style-type: none"> • Architected a WebSocket-based distributed simulation platform integrating a central controller, network-node workers, channel computation, mobility modeling, and visualization. • Designed priority-aware multi-channel access and cross-layer simulation logic covering traffic generation, routing, MAC access, physical transmission, interference, SINR, and reception decisions. • Built core algorithms and performance-evaluation workflows for large-scale scenarios involving more than 1,000 nodes. • Delivered a system prototype and one authorized software copyright. | <p>2022 – 2023</p> |
| <p>Institute of Computing Technology, Chinese Academy of Sciences, Core Contributor, Knowledge-Aware Cloud-Edge Orchestration for RAG Services</p> <ul style="list-style-type: none"> • Modeled heterogeneous knowledge bases, real-time computing loads, communication conditions, retrieval cost, generation quality, service delay, and energy consumption. • Designed a dynamic orchestration mechanism based on task semantics, node knowledge coverage, and resource states. • Built a cloud-edge RAG simulation environment for comparative evaluation. | <p>2025 – 2026</p> |

Institute of Computing Technology, Chinese Academy of Sciences , Core Contributor, Memory-Aware Distributed Routing for Space AI Computing Networks - Modeled the coupling among task routing, node memory state, computing load, storage availability, and link conditions. - Designed a distributed routing method for high-dimensional and dynamically changing network states. - Evaluated the approach using a ground-based space-computing-network simulation environment.	2025 – 2026
Institute of Computing Technology, Chinese Academy of Sciences , Lead Researcher and First Author, ISAC-Driven AIGC Service Orchestration - Designed the content-accuracy-and-quality-aware metric and modeled sensing-computing-communication coupling for AIGC services. - Formulated the joint resource-allocation problem and designed an LP-guided hierarchical deep reinforcement learning framework. - Implemented comparative simulations, ablation studies, and performance analysis.	2024 – 2025
Institute of Computing Technology, Chinese Academy of Sciences , Task Leader, Technical Lead, and First Author, Deception-Based UAV-Assisted Secure Communications - Coordinated the technical route for secure communication modeling, deception signaling, UAV three-dimensional placement, beamforming, and power allocation. - Built simulation validation workflows and delivered paper and invention-patent outputs related to anti-interference and secure communication. - Achieved a gap of no more than 3% from the global optimum and improved secure transmission rate by more than 100% over the no-deception scheme.	2021 – 2023

Publications

Service Satisfaction Based User Selection and Resource Allocation for NOMA-Based Multi-Cell MEC Networks Ningzhe Shi , Yiqing Zhou, Ling Liu, Yihao Wu, Hanxiao Yu, Jinglin Shi 10.1109/TMC.2026.3673357 (IEEE Transactions on Mobile Computing)	2026
Content Accuracy and Quality Aware Resource Allocation Based on LP-Guided DRL for ISAC-Driven AIGC Networks Ningzhe Shi , Yiqing Zhou, Ling Liu, Jinglin Shi, Yihao Wu, Haiwei Shi, Hanxiao Yu 10.1109/TMC.2026.3652144 (IEEE Transactions on Mobile Computing)	2026
Long-Term Fairness From Real-Time Decisions: Reward Adaptive Multi-Agent DRL for Wireless Communication With Imperfect CSI Ningzhe Shi , Ling Liu, Yiqing Zhou, Hanxiao Yu, Yihao Wu, Jingya Yang, Jinglin Shi 10.1109/TCCN.2026.3692166 (IEEE Transactions on Cognitive Communications and Networking)	2026
Deception-Based Secure Communication for UAV-Assisted mmWave MIMO Systems Ningzhe Shi , Yiqing Zhou, Yu Zhang, Zhijun Han, Ling Liu, Jinglin Shi 10.23919/JCC.fa.2025-0337.202604 (China Communications)	2026

Intellectual Property

抗跟踪式干扰的信号发送方法、信号接收方法及系统 Granted invention patent: CN202211134854.X	Sept 2024
一种基于中继位置选择和功率分配的抗干扰方法及系统 Granted invention patent: CN202310078808.0	May 2025

无线网络资源分配系统的构建方法和资源管理方法 Granted invention patent: CN202310354794.0	Apr 2026
无线网络资源分配系统的构建方法和资源管理方法 Patent application: PCT/CN2023/090283	Apr 2023
一种全双工自组织网络实时仿真系统及方法 Patent application: CN202311719290.0	Dec 2023
一种多用户 MIMO 信号检测模型 Patent application: CN202410018551.4	Jan 2024
一种面向自组织网络的信道接入方法和通信系统 Patent application: CN202410348337.5	Mar 2024
一种移动自组网数据路由方法、工业自组网系统 Patent application: CN202510873622.3	June 2025
一种环境和资源状态感知的资源分配方法和系统 Patent application: CN202510955830.8	July 2025
基于集成感知与通信的 AI 生成内容服务的资源分配方法 Patent application: CN202511053332.0	July 2025
联合信道估计与信号检测的方法、信号接收机及存储介质 Patent application: CN202511536493.5	Oct 2025
一种面向高速移动通信的发射装置及语义传输方法 Patent application: CN202610406457.5	Mar 2026
大规模全双工网络仿真系统 Registered software copyright: 2024SR0082117	Jan 2024

Honors and Awards

Outstanding Graduate Student, Xinghang Program Journal of Electronics & Information Technology; national selection of 22 graduate students	June 2026
Outstanding Poster Award Frontiers in Electronics and Information Conference	June 2026
UCAS Outstanding Student Model University-wide top 1%	June 2026
ICT Outstanding Student Institute of Computing Technology / State Key Laboratory of Processors	Jan 2026
E Fund Financial Technology Doctoral Scholarship 12 recipients institute-wide	Jan 2025
UCAS Outstanding Student Cadre	May 2025
UCAS Outstanding Student	June 2023
UCAS First-Class Scholarship Awarded for five consecutive years	2021–2025

Outstanding Graduate, Xidian University	June 2021
First-Class Graduation Scholarship, Xidian University	June 2021
Honorable Mention, Mathematical Contest in Modeling	2020
National Scholarship	2019
National Second Prize, China Undergraduate Mathematical Contest in Modeling	2019
Third Prize, CETC-10 Communication Cup	2019

Skills

Modeling and Optimization

Wireless and computing resource scheduling; ISAC; AI-service orchestration; convex optimization; linear programming; heuristic optimization

Artificial Intelligence

Deep reinforcement learning; multi-agent reinforcement learning; generative models; diffusion models; quantum reinforcement learning

Programming and Simulation

Python; C++; MATLAB; Shell; PyTorch; custom reinforcement-learning environments; system-level network simulation

Academic Writing

IEEE journal writing; technical reports; literature review; simulation-result analysis

Academic Service

Program Committee

- Technical Program Committee Member, IEEE International Conference on Communications (IEEE ICC) 2026

Academic Correspondence

- Academic Correspondent, Digital Communications and Networks

Peer Review

- IEEE Transactions on Mobile Computing
- IEEE Transactions on Communications
- IEEE Internet of Things Journal
- IEEE Transactions on Vehicular Technology
- Science China Information Sciences
- China Communications
- IEEE International Conference on Communications
- IEEE Vehicular Technology Conference
- 通信学报
- 电子与信息学报

Outreach and Public Engagement

Science outreach volunteer, Chinese Academy of Sciences Public Science Day

Participated in public science communication and research outreach activities

Committee Member, UCAS Science Popularization Association

Leadership

Vice President, Haidian Volunteer Federation Branch

Youth League Branch Secretary and Volunteer Team Leader, Xidian University School of Telecommunications Engineering